Data Wrangling II

import the requried libraries

```
import pandas as pd
```

Importing the dataset

```
data = pd.read_csv('xAPI-Edu-Data.csv')
data.head(10)
```

	gender	NationalITy	PlaceofBirth	StageID	GradeID	SectionID
0	M	KW	KuwaIT	lowerlevel	G-04	A
1	M	KW	KuwaIT	lowerlevel	G-04	A
2	M	KW	KuwaIT	lowerlevel	G-04	A
3	M	KW	KuwaIT	lowerlevel	G-04	A
4	M	KW	KuwaIT	lowerlevel	G-04	A
5	F	KW	KuwaIT	lowerlevel	G-04	A
6	M	KW	KuwaIT	MiddleSchool	G-07	A
7	M	KW	KuwaIT	MiddleSchool	G-07	A
8	F	KW	KuwaIT	MiddleSchool	G-07	A
9	F	KW	KuwaIT	MiddleSchool	G-07	B

Perfroming the operations on dataset

```
data.shape
```

```
(480, 17)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 480 entries, 0 to 479
Data columns (total 17 columns):
 #   Column                    Non-Null Count  Dtype 
---  ------                    --------------  ----- 
 0   gender                    480 non-null    object
 1   NationalITy               480 non-null    object
 2   PlaceofBirth              480 non-null    object
 3   StageID                   480 non-null    object
 4   GradeID                   480 non-null    object
 5   SectionID                 480 non-null    object
 6   Topic                     480 non-null    object
 7   Semester                  480 non-null    object
 8   Relation                  480 non-null    object
 9   raisedhands               480 non-null    int64 
 10  VisITedResources          480 non-null    int64 
 11  AnnouncementsView         480 non-null    int64 
 12  Discussion                480 non-null    int64 
 13  ParentAnsweringSurvey     480 non-null    object
 14  ParentschoolSatisfaction  480 non-null    object
 15  StudentAbsenceDays        480 non-null    object
 16  Class                     480 non-null    object
dtypes: int64(4), object(13)
memory usage: 63.9+ KB
```

```
# Int type columns are -raisedhands, VisITedResources, Announcement
data[['raisedhands', 'VisITedResources', 'AnnouncementsView']]
```

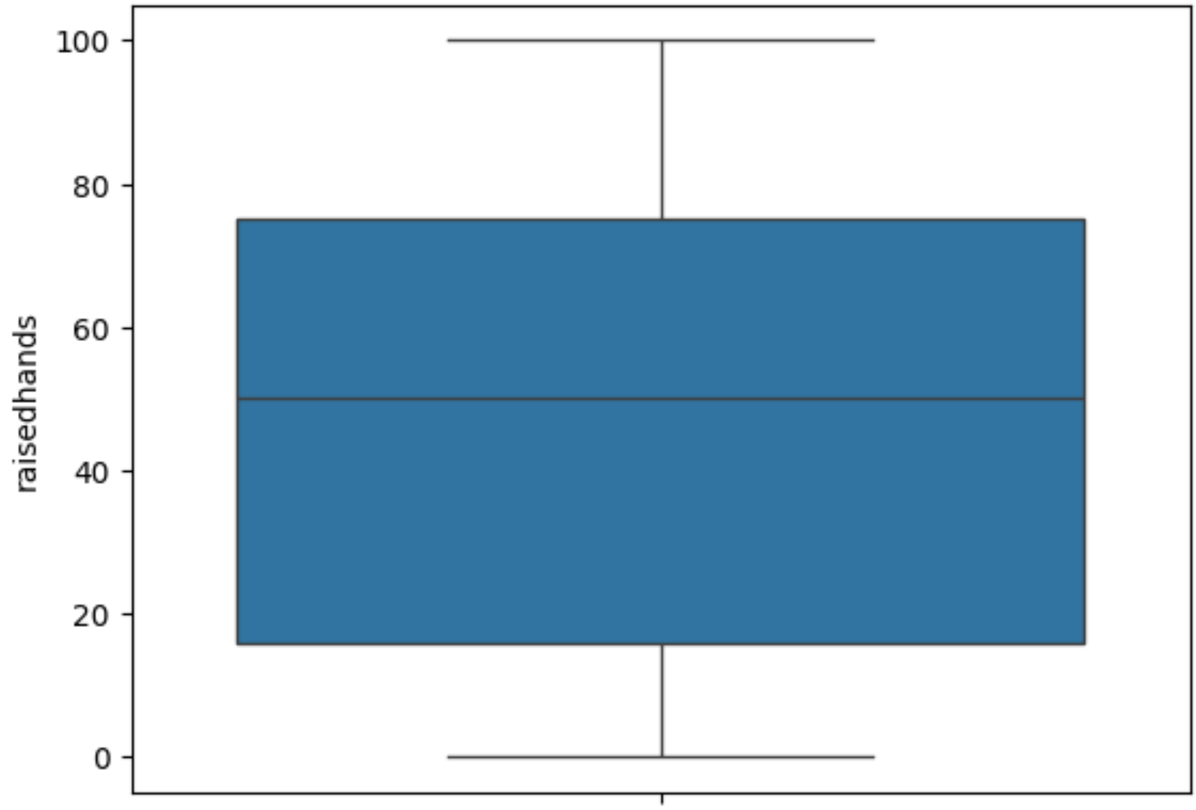
	raisedhands	VisITedResources	AnnouncementsView
0	15	16	2
1	20	20	3
2	10	7	0
3	30	25	5
4	40	50	12
...	...	...	...
475	5	4	5
476	50	77	14
477	55	74	25
478	30	17	14
479	35	14	23

480 rows × 3 columns

```
# Identify (Detect the outliers) with the help of BoxPlot
import seaborn as sns
```

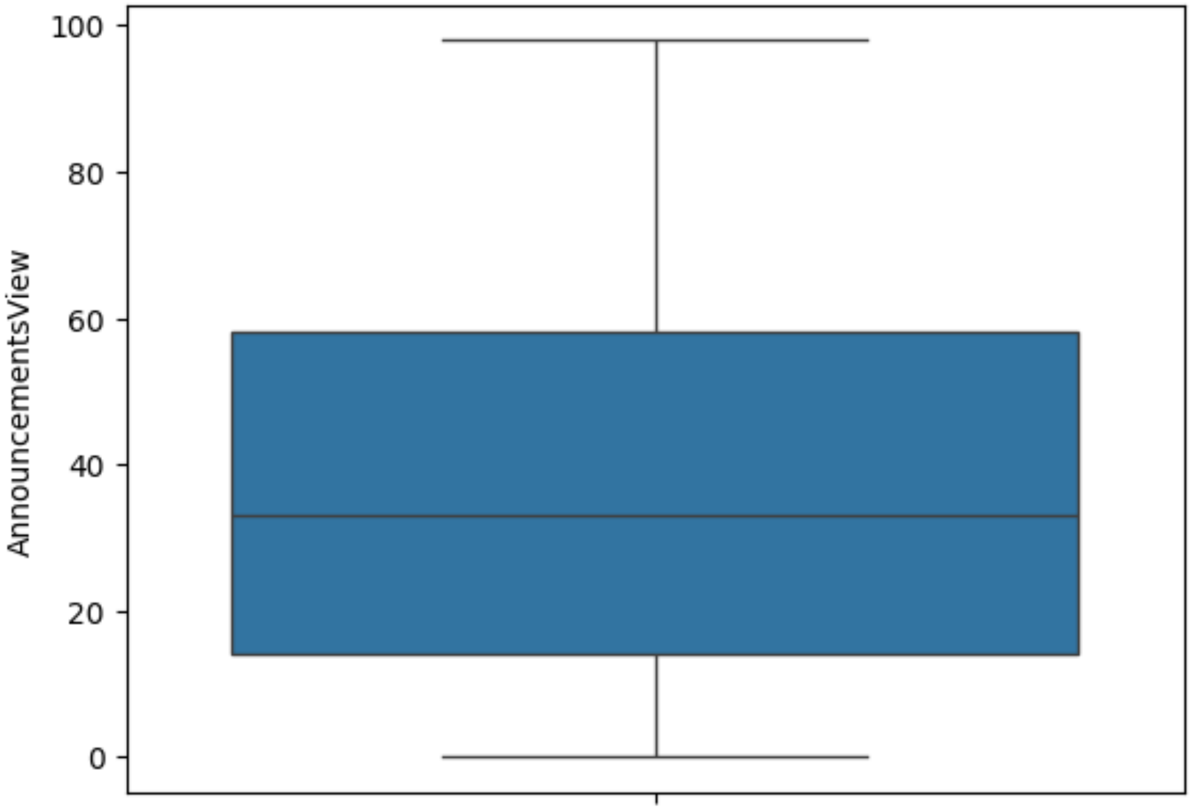
```
sns.boxplot(data['raisedhands'])
```

```
<Axes: ylabel='raisedhands'>
```



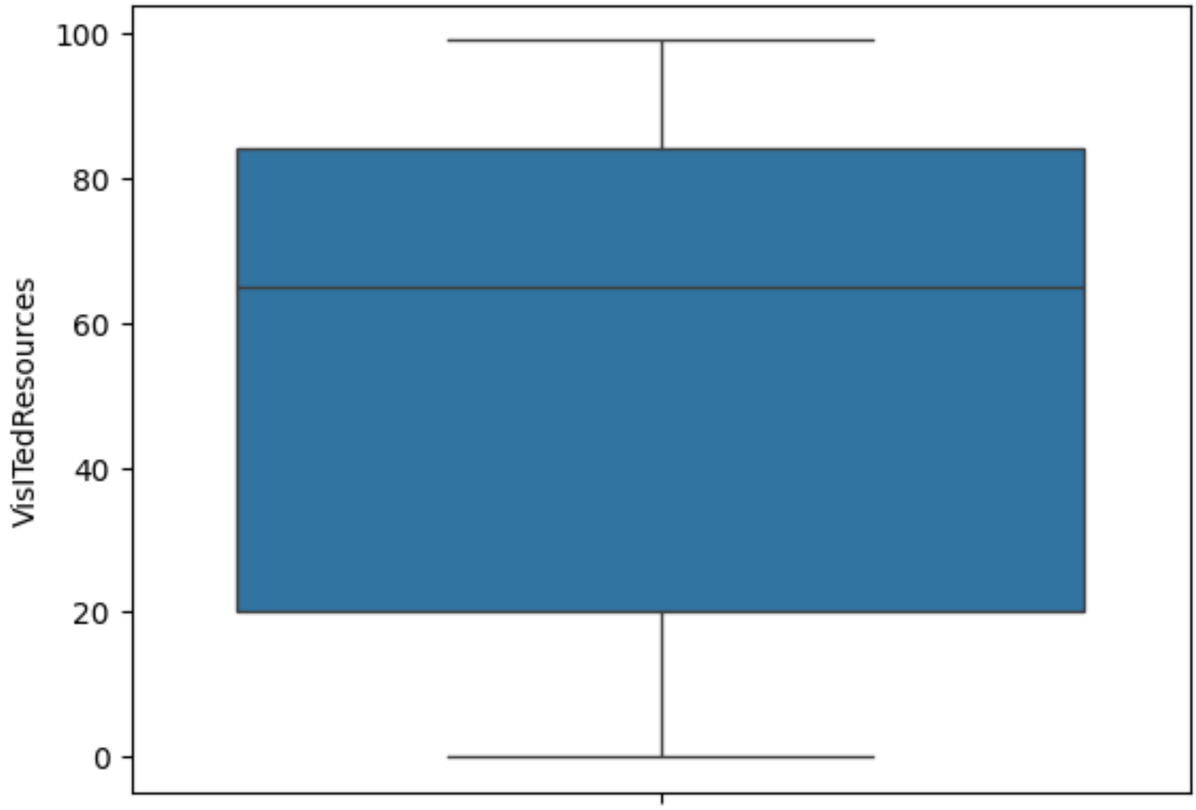
```
sns.boxplot(data['AnnouncementsView'])
```

```
<Axes: ylabel='AnnouncementsView'>
```



```
sns.boxplot(data['VisITedResources'])
```

<Axes: ylabel='VisITedResources'>



in all the three diagrams above there is not outlier